

PROJECT PROPOSAL: CENTRALIZED DEVELOPMENT & TESTING SERVER

Proposed by: TechOps Department	Target Audience: Pinc Technology Management
Supported by: 1. Zulhisyam Tan (HOD Backend) 2. Sharil (HOD Front End)	Proposed Budget: RM 9,000 – RM 12,000

1. Executive Summary

This proposal outlines the strategic acquisition of a high-performance physical server for Pinc Technology. With an approved budget of RM 9,000 to RM 12,000, we plan to purchase a refurbished enterprise-grade rack server to serve as our core local testing, sandbox, and development host.

Currently, hosting UAT, staging, client sandboxes, and developer sandbox databases on cloud providers (like AWS) incurs high, compounding monthly charges. Because development environments must run continuously but are often idle outside business hours, cloud utility pricing is highly inefficient for this workload. By moving our dev/test pipeline to a centralized on-premises or collocated server, we can eliminate recurring hosting costs, provide developers with massive sandbox environments, and deliver isolated, professional preview links to clients. Refurbished enterprise servers offer an outstanding 85% of modern performance for less than 15% of their original retail cost, representing an exceptional, risk-free ROI.

2. Server Use Cases & Business Justification

To ensure clear understanding and alignment, we have categorized the server's four primary workloads and their corresponding business values below:

- API Host for Dev & UAT Environments:** Backend and frontend developers require a hosting server to test API endpoints in the development and UAT (User Acceptance Testing) stages before push to production. Hosting these locally saves thousands in hourly cloud runtime fees.
- Client Sandbox Environments:** We regularly need to provide corporate clients with a staging site (or 'sandbox') to preview new features and sign off on project phases. Having our own host allows us to deploy, reset, and tear down client sandboxes instantly at zero extra cost.
- Developer Local Dev Playgrounds:** Setting up resource-intensive dependencies (such as large databases, Elasticsearch, or caching layers) directly on developers' laptops slows down their local systems. This server will host dedicated database and app sandboxes that developers can connect to remotely, boosting efficiency.

- **Open-Source Staff Applications:** We will run internal productivity tools that benefit all staff, such as self-hosted document portals (Nextcloud), team knowledge bases (Wiki.js / Bookstack), and internal task runners to automate repetitive workflows.

3. Technical Requirements

Because this server will host multiple operating systems and applications in virtual environments simultaneously, it must meet stringent resource thresholds to avoid bottlenecks:

- **Memory (128GB ECC RAM Minimum):** ECC (Error-Correcting Code) memory is non-negotiable. It automatically detects and corrects memory data errors in real-time, preventing system crashes and data corruption under high virtual machine density.
- **Processing Power:** Dual physical processors with high cores and threads are required. This ensures we can assign dedicated vCPUs to virtual machines so that a heavy task on one VM does not slow down other developers.
- **Enterprise Storage:** Enterprise-grade SSDs configured in hardware RAID (Redundant Array of Independent Disks) are essential to provide high-speed database reads/writes and ensure data remains safe even if a physical drive fails.

4. Server Model Detailed Analysis (Models 1 - 5)

We audited the five refurbished hardware packages available. Below is our evaluation of their specifications, age, hardware relevance, and original price when new, along with references of the server units:

Model 1: IBM System x3550 M5 (1U Rack Form Factor)

- Price (Refurbished): RM 9,094.24
- CPU: 2x Intel Xeon E5-2699 v3 (18 Cores, 36 Threads each | Total: 36 Cores, 72 Threads)
- RAM: 128GB DDR4 ECC | Storage: 8x 480GB SSD (3.84TB Raw)
- RAID Controller: ServeRAID M5210 | Power Supply: 750W
- Age of Model: Launched Q4 2014 (~11.5 years old).
- Relevance & Sufficiency: High. The Xeon E5-2699 v3 is a high-core beast. 72 threads allow high VM density. The 8x SSD package provides massive high-speed storage, ideal for virtualization.
- Price New (with Proof): Xeon E5-2699 v3 launched at a recommended customer price of \$4,115.00 USD each (~RM 18,106.00). Dual CPUs alone cost \$8,230.00 USD (~RM 36,212.00) (Source: Intel ARK Official CPU database). A fully loaded IBM x3550 M5 configured new with these specs retailed for approx. \$18,000.00 USD (approx. RM 79,200.00).



Figure 1: Front view of the IBM System x3550 M5 (1U Form Factor)

Model 2: Precomp Fusion 2288H V5 (2U Rack Form Factor)

- Price (Refurbished): RM 9,410.50
- CPU: 2x Intel Xeon Silver 4116 (12 Cores, 24 Threads each | Total: 24 Cores, 48 Threads)
- RAM: 128GB DDR4 ECC | Storage: 3x 480GB SSD (1.44TB Raw)
- RAID Controller: LSI SAS 3108 PCIe | Power Supply: 500W
- Age of Model: Launched Q3 2017 (~9 years old).
- Relevance & Sufficiency: Medium. Newer Skylake architecture has better energy efficiency (85W TDP) and instructions, but fewer cores (24 vs 36) and restricted storage (only 3 SSDs) make this less suited for running multiple environments.
- Price New (with Proof): Xeon Silver 4116 CPU launched at a recommended price of \$1,002.00 USD each (~RM 4,408.00) (Source: Intel ARK Specifications). A configured Huawei 2288H V5 server with these specs retailed for approx. \$9,500.00 USD (approx. RM 41,800.00).



Figure 2: Front view of the Precomp Fusion 2288H V5 (Huawei Base)

Model 3: Dell PowerEdge R630 (1U Rack Form Factor)

- Price (Refurbished): RM 9,323.50
- CPU: 2x Intel Xeon E5-2699 v3 (18 Cores, 36 Threads each | Total: 36 Cores, 72 Threads)
- RAM: 128GB DDR4 ECC | Storage: 8x 480GB SSD (3.84TB Raw)

- RAID Controller: Dell PERC H730 (12Gbps) | Power Supply: 750W
- Age of Model: Launched September 2014 (~11.5 years old).
- Relevance & Sufficiency: Very High. Dell PowerEdge servers are highly reliable, and replacement parts are extremely easy to acquire in Malaysia. The H730 RAID controller with dedicated cache provides superior IOPS. 72 threads and 8 SSDs offer the best virtualization density.
- Price New (with Proof): Xeon E5-2699 v3 launched at a recommended price of \$4,115.00 USD each (Total \$8,230.00 USD for dual CPUs) (Source: Intel ARK). A configured Dell R630 with 128GB RAM and 8 SSDs was priced at approx. \$19,500.00 USD (~RM 85,800.00) when new.



Figure 3: Front panel of the Dell PowerEdge R630 (1U Enterprise System)

Model 4: HPE ProLiant DL360 Gen10 (1U Rack Form Factor)

- Price (Refurbished): RM 9,142.50
- CPU: 2x Intel Xeon Silver 4116 (12 Cores, 24 Threads each | Total: 24 Cores, 48 Threads)
- RAM: 128GB DDR4 ECC | Storage: 3x 480GB SSD (1.44TB Raw)
- RAID Controller: HPE Smart Array P408i | Power Supply: 500W
- Age of Model: Launched Q3 2017 (~9 years old).
- Relevance & Sufficiency: High (for modernization). The Gen10 platform is highly secure and features HPE iLO 5 for remote management. However, 3 SSDs are restrictive; RAID 5 leaves only 960GB usable storage, requiring a storage upgrade soon.
- Price New (with Proof): Xeon Silver 4116 CPU launched at \$1,002.00 USD each (Source: Intel ARK). An HPE DL360 Gen10 configured with these specs retailed at around \$9,800.00 USD (approx. RM 43,120.00) in 2017.



Figure 4: Front panel of the HPE ProLiant DL360 Gen10 (1U Modern System)

Model 5: IBM System x3650 M5 (2U Rack Form Factor)

- Price (Refurbished): RM 9,584.50
- CPU: 2x Intel Xeon E5-2699 v3 (18 Cores, 36 Threads each | Total: 36 Cores, 72 Threads)
- RAM: 128GB DDR4 ECC | Storage: 8x 480GB SSD (3.84TB Raw)
- RAID Controller: ServeRAID M5210 | Power Supply: 750W
- Age of Model: Launched Q4 2014 (~11.5 years old).
- Relevance & Sufficiency: Very High. As a 2U rack server, it provides superior cooling, operates much more quietly than 1U servers (ideal for offices), and has plenty of PCIe slots and drive bays for future expansion.
- Price New (with Proof): Xeon E5-2699 v3 launched at \$4,115.00 USD each (Source: Intel ARK). An IBM x3650 M5 configured new with these specs retailed for approx. \$20,000.00 USD (approx. RM 88,000.00).



Figure 5: IBM System x3650 M5 Chassis Design (Note: 2U model shares identical design aesthetics with the 1U x3650 M5 but has double the physical height and higher bay density)

5. Comprehensive Comparison Table

Specification / Metric	Model 1: IBM x3550 M5	Model 2: Precomp Fusion	Model 3: Dell R630	Model 4: HPE DL360 G10	Model 5: IBM x3650 M5
Form Factor	1U Rack	2U Rack	1U Rack	1U Rack	2U Rack
Total CPU Cores	36 Cores	24 Cores	36 Cores	24 Cores	36 Cores
Total CPU Threads	72 Threads	48 Threads	72 Threads	48 Threads	72 Threads
ECC Memory	128GB DDR4	128GB DDR4	128GB DDR4	128GB DDR4	128GB DDR4
Included Storage	8 x 480GB SSD (3.84TB)	3 x 480GB SSD (1.44TB)	8 x 480GB SSD (3.84TB)	3 x 480GB SSD (1.44TB)	8 x 480GB SSD (3.84TB)
Hardware RAID	ServeRAID M5210	LSI SAS 3108	Dell PERC H730	HPE Smart Array P408i	ServeRAID M5210
Power Supply	750W	500W	750W	500W	750W
Age of Model	~11.5 Years (2014)	~9 Years (2017)	~11.5 Years (2014)	~9 Years (2017)	~11.5 Years (2014)
Price New (Est.)	RM 79,200.00	RM 41,800.00	RM 85,800.00	RM 43,120.00	RM 88,000.00
Refurbished Price	RM 9,094.24	RM 9,410.50	RM 9,323.50	RM 9,142.50	RM 9,584.50
Value Rating	Good	Fair	Excellent	Very Good	Excellent

6. Strategic Recommendations

Based on TechOps' auditing, we recommend the following purchase paths based on our deployment layout:

Option A: Model 5 — IBM System x3650 M5 (2U Rack) — [Top Overall Pick]

- Why: Running multiple virtual machines and databases creates heavy CPU heat. The 2U rack form factor has larger, slower-moving fans. This makes the server run significantly quieter than 1U models (critical if kept in the office) and keeps components cooler for long-term health.
- Key Advantage: It includes the high-capacity 36 Cores / 72 Threads and 8 x 480GB SSDs (3.84TB raw). Setting these drives up in a RAID 10 configuration provides 1.92TB of high-speed, fully redundant storage. The 2U chassis also leaves physical space to double the memory or add more drives as Pinc's demands grow.
- Refurbished Price: RM 9,584.50 (Comfortably within the RM9,000 - RM12,000 budget).

Option B: Model 3 — Dell PowerEdge R630 (1U Rack) — [Top Pick for Server Racks]

- Why: If Pinc plans to put this server in a standard server rack or host it in a colocation facility, the

Dell PowerEdge R630 is the gold standard.

- **Key Advantage:** Dell parts are highly accessible in Malaysia, and Dell's iDRAC 8 remote management is widely supported. It shares the same high-end specifications as Model 5 (36 Cores, 128GB RAM, 8 SSDs) in a compact 1U form factor. The Dell PERC H730 is a superior hardware RAID card.
- **Refurbished Price:** RM 9,323.50 (Outstanding price-to-performance ratio).

7. Hypervisor & Software Architecture

To maximize the utility of the 128GB RAM and 36 Cores, TechOps will deploy Proxmox VE (a free, open-source hypervisor) directly on the hardware. This allows us to partition the physical hardware into isolated virtual containers running Ubuntu Linux, divided as follows:

- **API Host VM (Dev/UAT):** 4 vCPUs, 32GB RAM, 500GB Storage (Runs Docker & Portainer to deploy APIs quickly)
- **Client Sandboxes VM:** 8 vCPUs, 32GB RAM, 500GB Storage (Isolated preview environment for client projects)
- **Developer Dev VM:** 8 vCPUs, 32GB RAM, 500GB Storage (Shared databases and services for frontend/backend devs)
- **Open-Source Staff VM:** 4 vCPUs, 16GB RAM, 300GB Storage (For Nextcloud file shares, Wiki.js document portals, etc.)
- **System Buffer (Unallocated):** 12 vCPUs, 16GB RAM (Kept unallocated to ensure host stability and absorb traffic spikes)

8. Conclusion

By investing between RM 9,300 to RM 9,600 once in a refurbished enterprise server, Pinc Technology will build a permanent development infrastructure that cuts cloud billing, boosts developer throughput, and provides professional staging systems for our clients.

We request the management to consider and approve this purchase of either Model 5 (IBM x3650 M5 2U) or Model 3 (Dell PowerEdge R630 1U) so that TechOps can proceed with ordering, OS deployment, and networking setup.